

Technical Document #1019: Machine Learning - Comprehensive Overview

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The bias-variance tradeoff is a fundamental concept in machine learning. High bias leads to underfitting while high variance leads to overfitting.

Random forests are ensemble learning methods that construct multiple decision trees during training and output the class that is the mode of the classes of individual trees.

Neural networks are computing systems inspired by biological neural networks. They consist of layers of interconnected nodes that process information using connectionist approaches.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Support vector machines (SVMs) are supervised learning models that analyze data for classification and regression analysis.
- Feature selection is the process of selecting a subset of relevant features for use in model construction.
- Random forests are ensemble learning methods that construct multiple decision trees during training and output the class that